

## MANISTEE COUNTY BLACKER AP, MI, USA

WMO: 726385

Lat: 44.267N

Lon: 86.250W

Elev: 621

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94894

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 1.4        | 6.9 | -6.0                          | 4.1 | 4.9  | -1.5 | 5.2 | 8.6  | 25.4                  | 20.6 | 23.5 | 20.7 | 8.8                   | 90   | 0.564 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.2                   | 85.1            | 72.0 | 82.2 | 69.6 | 80.6 | 68.4 | 74.0                | 82.3 | 72.1 | 79.8 | 70.4 | 77.4 | 9.9                  | 210  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.7                            | 119.7 | 79.7 | 69.9 | 112.4 | 77.4 | 67.8 | 104.5 | 75.5 | 37.9          | 82.4 | 36.3 | 79.4 | 34.7 | 77.0 |                   |
|                                 |       |      |      |       |      |      |       |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 22.3              | 19.4 | 17.5 | -5.5                       | 89.8 | 6.6                | 2.5 | -10.3                                              | 91.5 | -14.1      | 93.0 | -17.8      | 94.4 | -22.6      | 96.2 |    |    |
|                   |      |      | -6.4                       | 76.8 | 6.5                | 2.1 | -11.1                                              | 78.2 | -14.9      | 79.5 | -18.6      | 80.6 | -23.3      | 82.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 45.7   | 23.5  | 24.6  | 31.6  | 42.9  | 53.9  | 62.8  | 67.2  | 66.1  | 59.6  | 48.4  | 37.7  | 28.9 |
|                                                                                  | DBStd       | 17.68  | 9.28  | 9.58  | 10.72 | 9.02  | 8.50  | 7.28  | 6.08  | 5.61  | 7.61  | 8.15  | 8.22  | 8.32 |
|                                                                                  | HDD50       | 3597   | 820   | 712   | 581   | 253   | 55    | 3     | 0     | 0     | 9     | 130   | 380   | 654  |
|                                                                                  | HDD65       | 7353   | 1285  | 1132  | 1035  | 665   | 359   | 124   | 48    | 55    | 195   | 517   | 820   | 1118 |
|                                                                                  | CDD50       | 2038   | 0     | 0     | 12    | 39    | 176   | 388   | 533   | 498   | 299   | 82    | 10    | 1    |
|                                                                                  | CDD65       | 318    | 0     | 0     | 1     | 1     | 15    | 59    | 116   | 88    | 34    | 4     | 0     | 0    |
|                                                                                  | CDH74       | 2698   | 0     | 0     | 7     | 16    | 171   | 565   | 1005  | 657   | 254   | 23    | 0     | 0    |
|                                                                                  | CDH80       | 570    | 0     | 0     | 1     | 1     | 34    | 133   | 234   | 123   | 43    | 1     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.7    | 9.1   | 8.9   | 8.3   | 8.8   | 7.6   | 6.4   | 5.9   | 5.8   | 6.4   | 7.9   | 8.8   | 8.8  |
| Precipitation                                                                    | PrecAvg     | 34.5   | 2.2   | 1.7   | 2.0   | 3.0   | 3.4   | 3.4   | 3.3   | 3.2   | 3.5   | 3.5   | 3.0   | 2.5  |
|                                                                                  | PrecMax     | 45.4   | 4.4   | 3.1   | 4.9   | 5.4   | 8.1   | 8.5   | 6.8   | 5.6   | 6.7   | 8.9   | 6.9   | 7.5  |
|                                                                                  | PrecMin     | 26.9   | 0.6   | 0.7   | 0.4   | 0.8   | 0.8   | 1.2   | 0.7   | 0.9   | 0.6   | 0.7   | 0.7   | 0.5  |
|                                                                                  | PrecStd     | 4.6    | 1.0   | 0.7   | 1.0   | 1.2   | 1.7   | 2.0   | 1.6   | 1.1   | 1.6   | 2.0   | 1.6   | 1.3  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 48.5  | 54.0  | 71.7  | 76.6  | 84.0  | 88.0  | 89.7  | 88.0  | 85.4  | 77.4  | 65.9  | 55.4 |
|                                                                                  |             | MCWB   | 44.5  | 47.9  | 58.6  | 60.7  | 68.9  | 73.3  | 75.2  | 73.6  | 72.5  | 66.0  | 57.0  | 51.3 |
|                                                                                  | 2%          | DB     | 44.5  | 46.4  | 61.2  | 70.3  | 79.2  | 83.8  | 85.0  | 82.9  | 80.8  | 72.0  | 59.4  | 47.8 |
|                                                                                  |             | MCWB   | 40.4  | 41.1  | 52.0  | 56.1  | 64.4  | 70.0  | 72.5  | 71.2  | 68.3  | 62.2  | 52.2  | 44.5 |
|                                                                                  | 5%          | DB     | 39.0  | 41.5  | 53.9  | 63.9  | 73.8  | 81.0  | 82.0  | 80.8  | 77.0  | 67.5  | 55.1  | 44.7 |
|                                                                                  |             | MCWB   | 36.0  | 37.0  | 46.7  | 51.6  | 61.3  | 67.5  | 70.3  | 69.4  | 66.7  | 59.7  | 49.0  | 40.6 |
|                                                                                  | 10%         | DB     | 36.5  | 37.2  | 46.5  | 58.7  | 70.2  | 77.0  | 79.5  | 77.8  | 73.1  | 63.0  | 51.7  | 39.6 |
|                                                                                  |             | MCWB   | 34.2  | 33.8  | 40.9  | 48.1  | 58.5  | 65.3  | 68.2  | 67.6  | 64.5  | 56.2  | 46.6  | 36.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 46.4  | 48.3  | 60.2  | 63.3  | 70.7  | 75.1  | 77.2  | 75.5  | 73.4  | 69.0  | 58.1  | 52.9 |
|                                                                                  |             | MCDB   | 47.5  | 50.2  | 69.4  | 73.1  | 81.3  | 85.1  | 86.7  | 82.8  | 81.7  | 74.3  | 63.5  | 55.3 |
|                                                                                  | 2%          | WB     | 40.2  | 41.8  | 53.5  | 58.0  | 67.0  | 71.6  | 74.7  | 73.2  | 70.7  | 64.6  | 54.1  | 45.5 |
|                                                                                  |             | MCDB   | 43.6  | 46.9  | 59.5  | 67.7  | 76.3  | 81.5  | 83.0  | 80.6  | 77.3  | 69.2  | 58.8  | 47.4 |
|                                                                                  | 5%          | WB     | 36.9  | 37.4  | 46.5  | 53.3  | 63.8  | 69.0  | 72.3  | 71.4  | 68.4  | 60.1  | 49.9  | 40.5 |
|                                                                                  |             | MCDB   | 39.1  | 41.0  | 51.8  | 61.9  | 71.8  | 77.6  | 79.9  | 77.9  | 74.6  | 66.5  | 54.8  | 44.0 |
|                                                                                  | 10%         | WB     | 34.0  | 34.3  | 41.5  | 49.4  | 60.2  | 67.1  | 70.3  | 69.5  | 66.1  | 56.5  | 46.2  | 37.1 |
|                                                                                  |             | MCDB   | 36.5  | 37.6  | 47.2  | 58.1  | 67.8  | 74.6  | 77.4  | 75.3  | 72.4  | 62.0  | 50.5  | 40.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.9  | 13.3  | 16.4  | 18.9  | 21.2  | 20.6  | 20.2  | 19.5  | 20.1  | 17.5  | 13.1  | 10.2 |
|                                                                                  |             | MCDBR  | 14.5  | 17.0  | 23.5  | 27.2  | 26.2  | 24.0  | 22.8  | 22.2  | 23.3  | 23.7  | 20.7  | 15.9 |
|                                                                                  |             | MCWBR  | 12.6  | 13.3  | 16.6  | 16.5  | 15.3  | 13.2  | 12.5  | 12.4  | 13.4  | 15.6  | 15.7  | 13.7 |
|                                                                                  | 5% WB       | MCDBR  | 13.6  | 15.5  | 21.5  | 24.5  | 22.7  | 20.6  | 19.4  | 18.3  | 19.3  | 20.1  | 19.0  | 15.1 |
|                                                                                  |             | MCWBR  | 12.3  | 13.0  | 16.5  | 16.8  | 15.0  | 12.6  | 11.5  | 11.2  | 12.4  | 15.1  | 15.5  | 13.7 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.292  | 0.307 | 0.329 | 0.360 | 0.390 | 0.404 | 0.405 | 0.397 | 0.379 | 0.346 | 0.314 | 0.286 |      |
|                                                                                  | taud        | 2.434  | 2.373 | 2.403 | 2.348 | 2.334 | 2.334 | 2.345 | 2.361 | 2.400 | 2.475 | 2.501 | 2.484 |      |
|                                                                                  | Ebn at Noon | 265    | 280   | 288   | 286   | 279   | 275   | 273   | 272   | 267   | 263   | 255   | 257   |      |
|                                                                                  | Edh at Noon | 23     | 29    | 32    | 37    | 39    | 39    | 38    | 36    | 32    | 26    | 22    | 20    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 347    | 630   | 1114  | 1480  | 1800  | 1971  | 1999  | 1695  | 1321  | 791   | 438   | 274   |      |
|                                                                                  | RadStd      | 39     | 70    | 119   | 139   | 155   | 134   | 88    | 107   | 117   | 91    | 64    | 35    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (47 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page